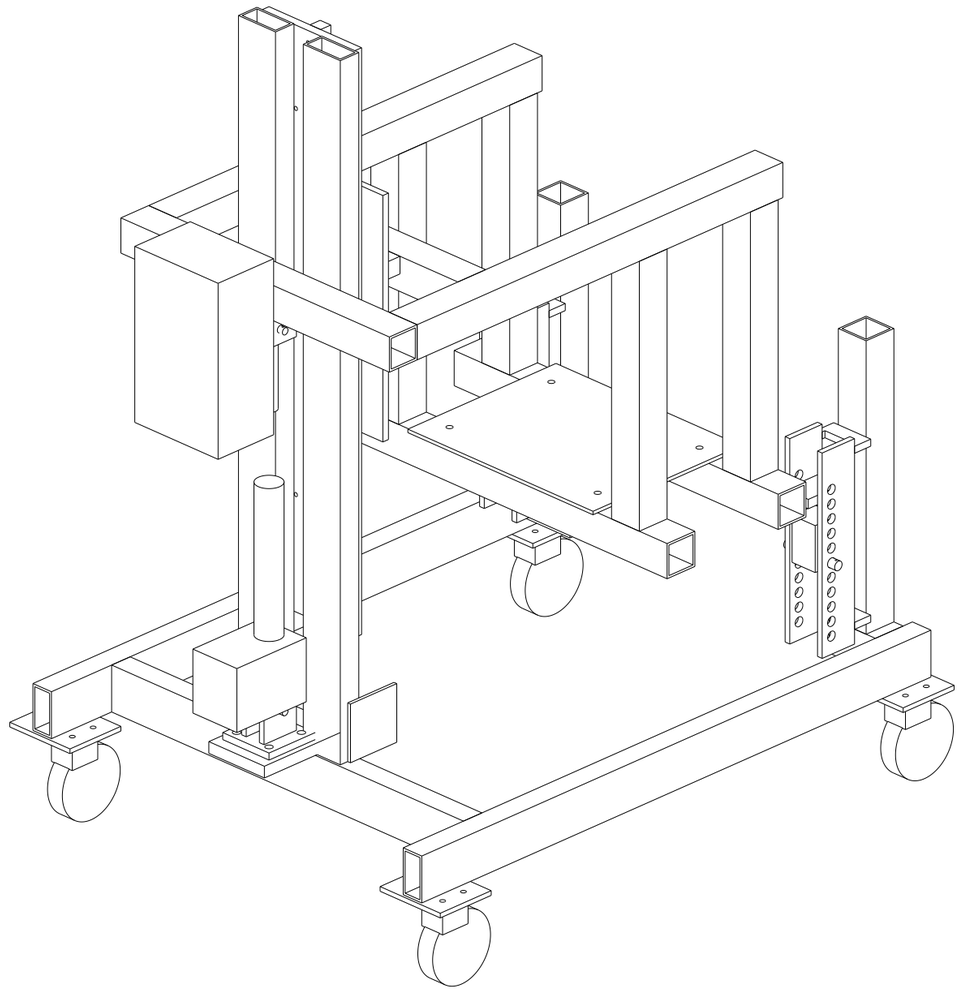
FreeStep P0-B1 / build review package

Dimensioned engineering review | secured ballast only



Purpose: test one guided lift, retained stationary perch and recovery with secured dummy ballast.

Geometry: 250 mm vertical travel; pan top 430 / 555 / 680 mm. Base caster axes 800 x 1000 mm. Steel frame top 1451 mm.

Mass: nominal custom steel 152.0 kg; moving custom steel 58.3 kg. Purchased parts, welds and guards excluded. Workshop fixture, not portable product.

Selected parts: LINAK LA40PL matched lift kit; HGR25 rail with two HGW25CC blocks; four Blickle 744589 total-lock casters.

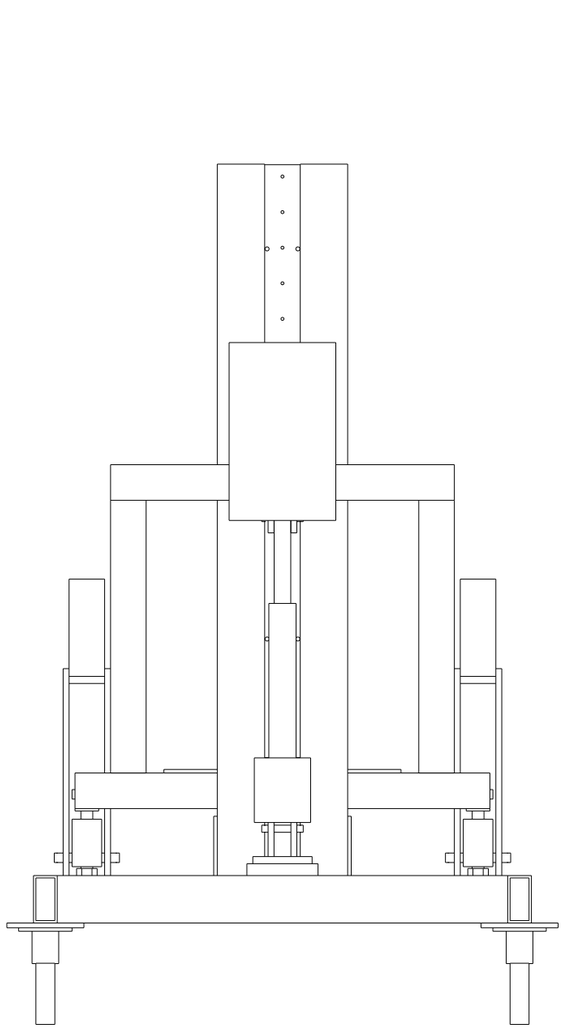
Status: supplier fit, structural joints, electrical interfaces, guarding and all physical testing remain open. See H01-H10.

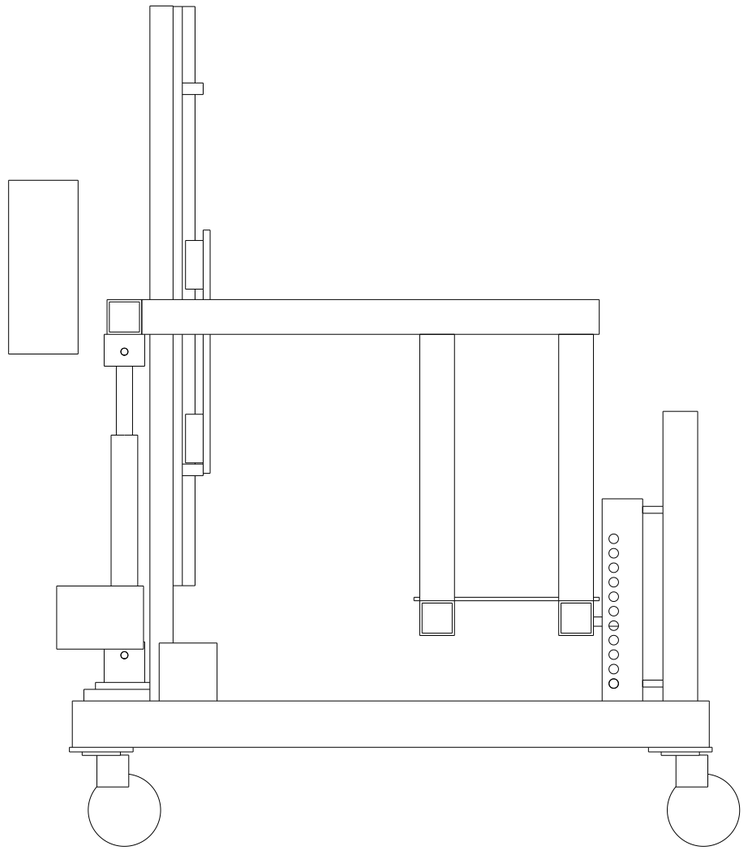
CAD contents: editable solid source, three STEP assembly poses, custom STEP solids, bored plate DXFs, cutlist and manifest. Grey purchased shapes are exclusion envelopes, not OEM CAD.

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

01 / General arrangement and coordinate datums

Dimensioned engineering review | secured ballast only





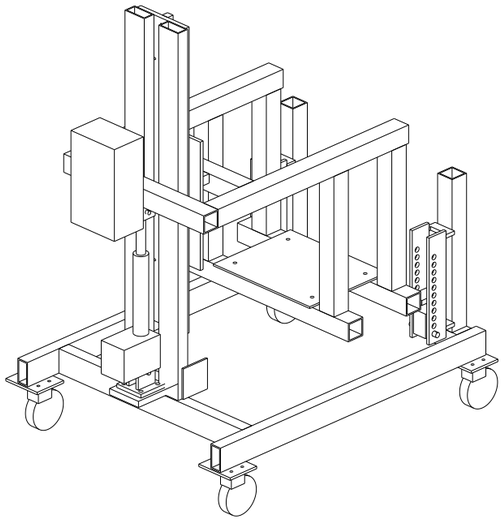
Datums: A = top of four caster plates, Z171. B = guide mounting face, Y140. C = X0 symmetry plane. Floor = Z0; +Y points to ballast. Castor axes X +/-400, Y0/1000.

Envelope: plates span930 mm X; base legs Y-50..1050; mast top1451. Wheel sweep/pedals and guards can enlarge it. Caster height163 mm provisional; newer catalogue164 mm is H03.

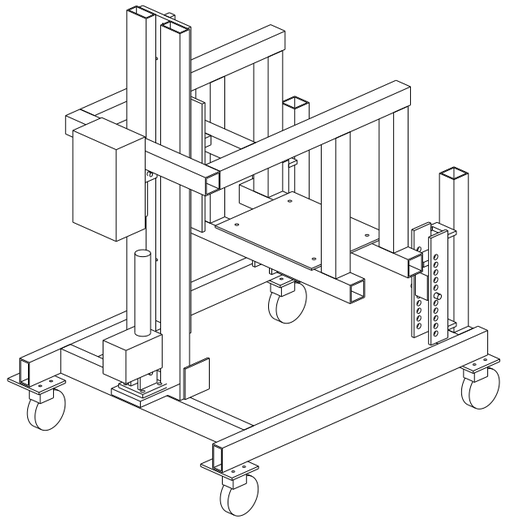
REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

02 / Travel, guide and actuator interfaces

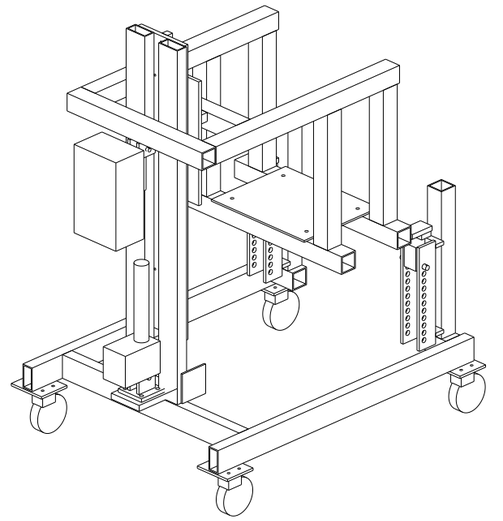
Dimensioned engineering review | secured ballast only



LOWERED



MID



RAISED

Condition	Lowered	Middle	Raised
Extension / pan top	0 / 430	125 / 555	250 / 680
Actuator upper pin Z	854	979	1104
Block centres Z	704 / 1004	829 / 1129	954 / 1254
Perch pin row Z	281	406	531

LA40PL assumption: lower pin (0,40,330), retracted pin distance524 = stroke250+274, upper pin rises vertically. Clevis gap28.5, PIN12 nominal12. Rail runs Z450..1450. Nominal250mm stroke; usable command travel may be shorter after limit/stop-distance calibration. Verify OEM drawing and all fit tolerances before machining.

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

03 / Perch props and controlled recovery

Dimensioned engineering review | secured ballast only

Feature	Dimension / function
Ladder P08	4 plates, paired wall gap60; PIN16 rows Z281+25n, n0..10; pin axis Y885
P15 screw block	50X x70Y x80Z; pin16.5 through X at local Y-15, Z15; M20x2.5 through at Y+15
Jackscrew / cap	ISO4017 M20x100-8.8 full thread; exposure17 nominal (14..20); cap40x40x20, underside pocket20.5 x3
Moving tab P09	50x85x16; world Y850..935, underside Z380+extension
P11 relief	Front edge relief60 wide x6 deep; cap rear edge935, fixed bridge material begins941 in central band
Lower-pose height chain	Pin281; P15 bottom266/top346; screw underhead263/tip363; cap360..380; tab underside380

Engage: stop at matching25 mm row, release controls, insert and retain both pins, adjust caps toward tabs. Either prop proof inhibits both electrical directions. Use approved OEM manual lowering <=1 mm to seat both.

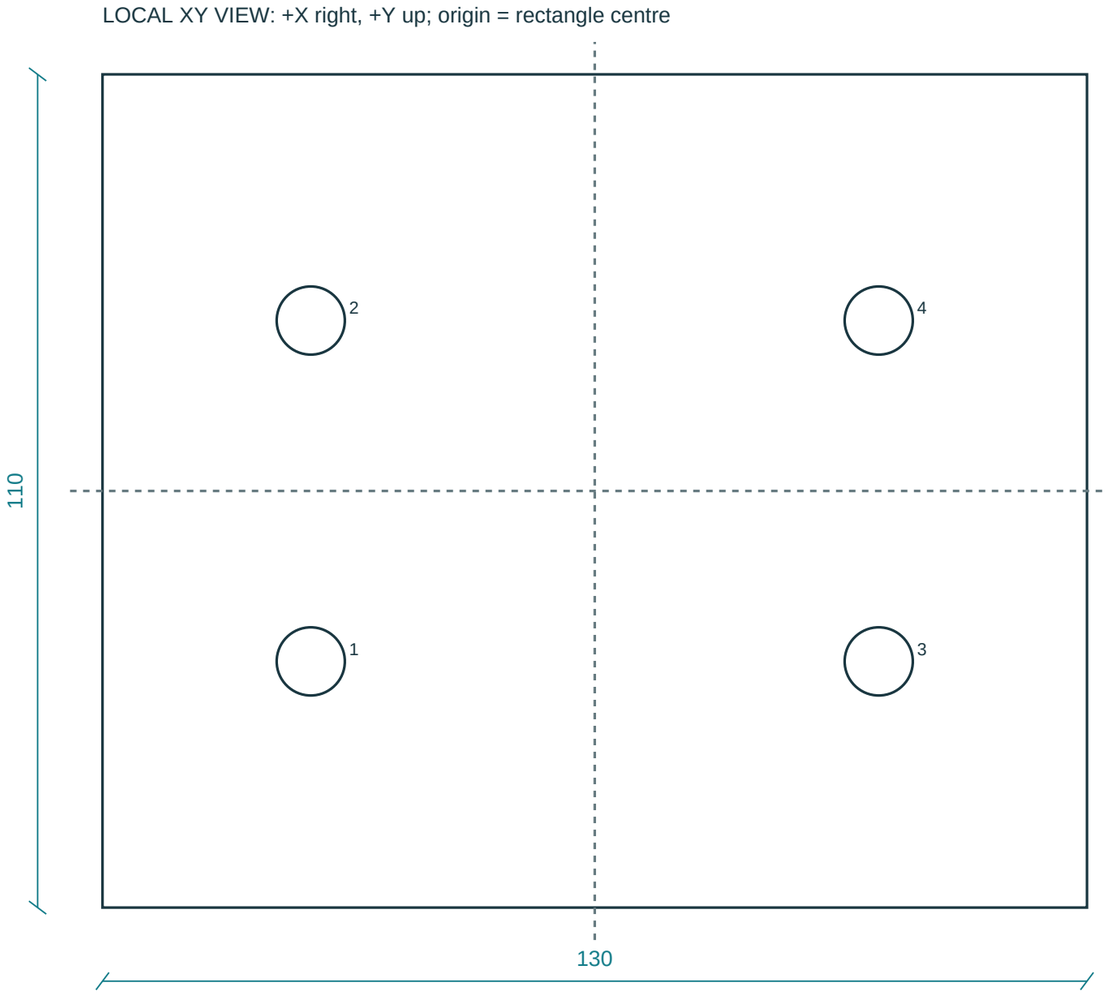
Release: keep OEM actuator connected and self-locking. Retract screw heads evenly in <=quarter-turn increments, <=3 mm total, until OEM support takes the load and pins are free. Remove only unloaded pins. No bypass of lock input.

Recovery: if pins remain loaded or a screw binds, isolate and use independently rated overhead recovery equipment. Never pry, pull a loaded pin or enter beneath the assembly.

Review: one prop, its tab weld, pin, ladder bridge and screw must be assessed for full moving load. Thread stripping, eccentric bearing, cap friction and fatigue are unverified.

P01 / Caster plate

Dimensioned engineering review | secured ballast only



Quantity: 4 | **Material:** S355 / YSt355; mill certificate required

Size: 130 x 110 x 8 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.882 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-37.5	-22.5	9 through
2	-37.5	22.5	9 through
3	37.5	-22.5	9 through
4	37.5	22.5	9 through

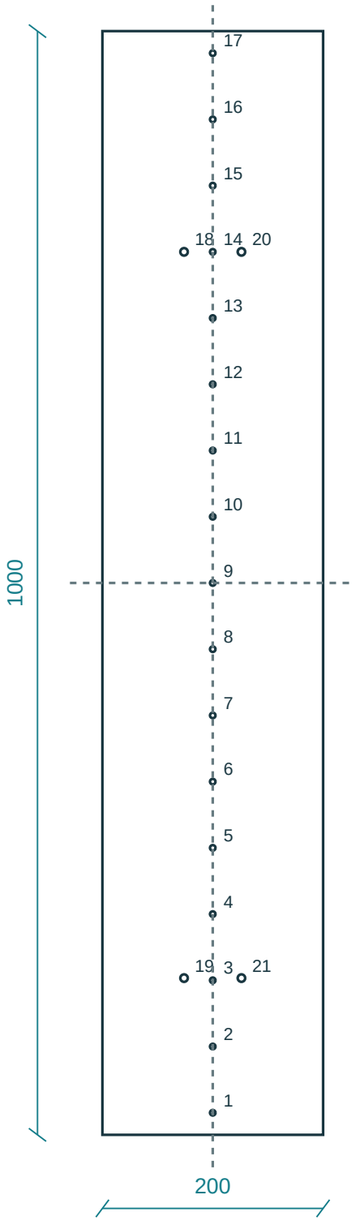
DXF: cad/generated/dxf/P01.dxf | STEP: cad/generated/parts/P01.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P02 / Mast backing plate

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



DXF: cad/generated/dxf/P02.dxf | STEP: cad/generated/parts/P02.step

Quantity: 1 | **Material:** S355 / YSt355; mill certificate required

Size: 200 x 1000 x 16 thick. Thickness extrudes local +Z.

Nominal unit mass: 25.060 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

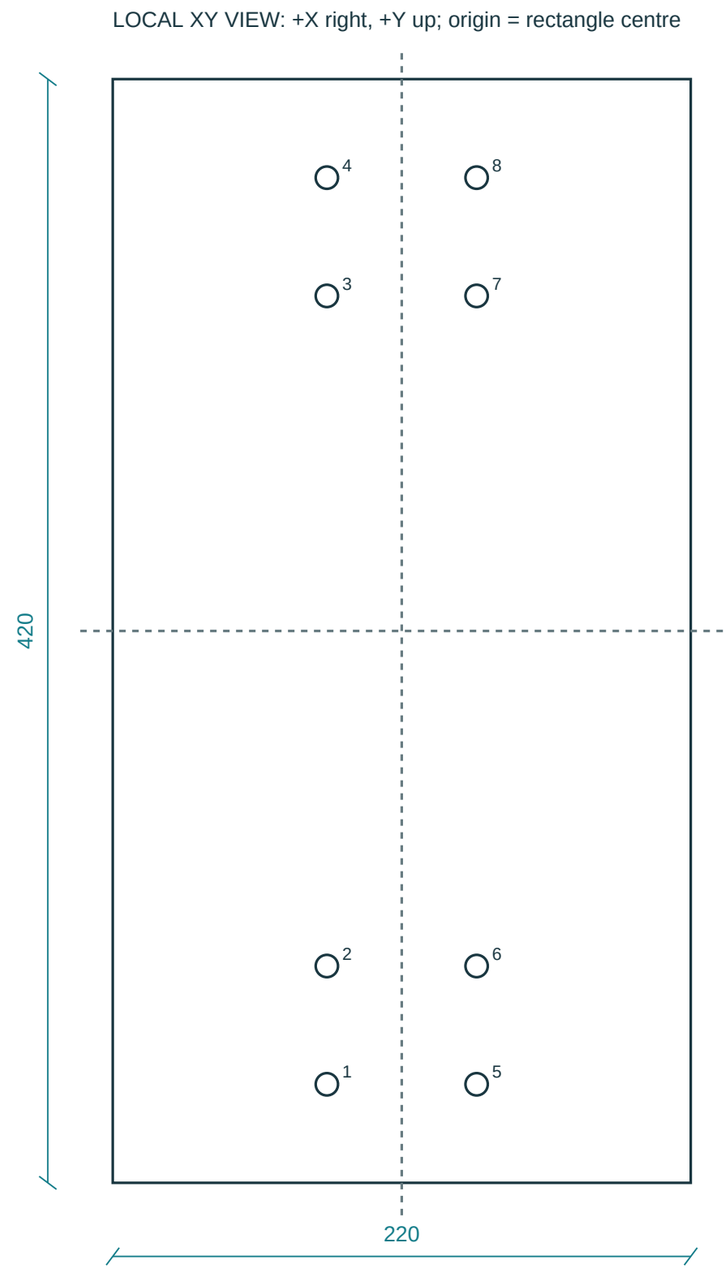
#	X	Y	Dia / operation
1	0	-480	M6 tap pilot5
2	0	-420	M6 tap pilot5
3	0	-360	M6 tap pilot5
4	0	-300	M6 tap pilot5
5	0	-240	M6 tap pilot5
6	0	-180	M6 tap pilot5
7	0	-120	M6 tap pilot5
8	0	-60	M6 tap pilot5
9	0	0	M6 tap pilot5
10	0	60	M6 tap pilot5
11	0	120	M6 tap pilot5
12	0	180	M6 tap pilot5
13	0	240	M6 tap pilot5
14	0	300	M6 tap pilot5
15	0	360	M6 tap pilot5
16	0	420	M6 tap pilot5
17	0	480	M6 tap pilot5
18	-26	300	M8 tap pilot6.8
19	-26	-358	M8 tap pilot6.8
20	26	300	M8 tap pilot6.8
21	26	-358	M8 tap pilot6.8

Machine datum B after welding. Local +Y becomes world -Z. World stop axes650/1308 correspond local +300/-358. Rail end distance must match received rail.

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P03 / Moving bridge

Dimensioned engineering review | secured ballast only



DXF: cad/generated/dxf/P03.dxf | STEP: cad/generated/parts/P03.step

Quantity: 1 | **Material:** S355 / YSt355; mill certificate required

Size: 220 x 420 x 12 thick. Thickness extrudes local +Z.

Nominal unit mass: 8.661 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-28.5	-172.5	8.5 through
2	-28.5	-127.5	8.5 through
3	-28.5	127.5	8.5 through
4	-28.5	172.5	8.5 through
5	28.5	-172.5	8.5 through
6	28.5	-127.5	8.5 through
7	28.5	127.5	8.5 through
8	28.5	172.5	8.5 through

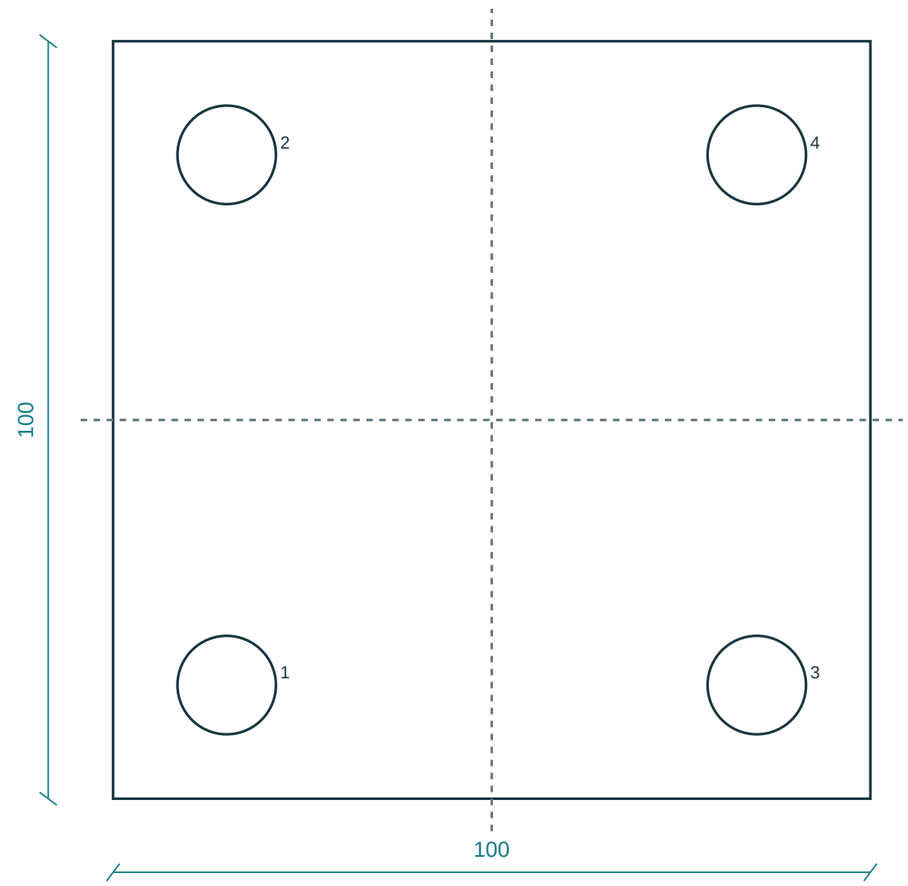
Mount two HGW25CC blocks at300 mm centres. Each57 x45 M8 pattern. Machine mating face after welding. Confirm OEM torque/thread depth.

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P04 / Lower clevis removable base

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



Quantity: 1 | **Material:** S355 / YSt355; mill certificate required

Size: 100 x 100 x 12 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.892 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-35	-35	13 through
2	-35	35	13 through
3	35	-35	13 through
4	35	35	13 through

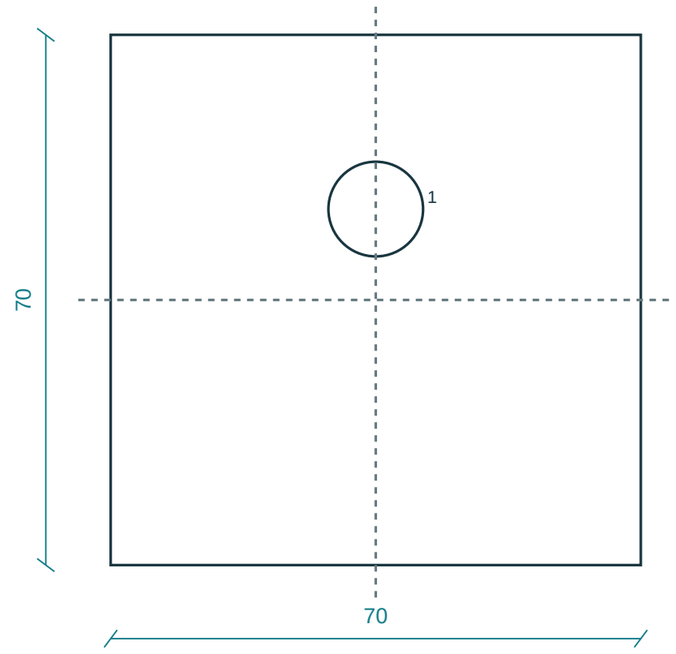
DXF: cad/generated/dxf/P04.dxf | STEP: cad/generated/parts/P04.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P05 / Lower clevis ear

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



Quantity: 2 | **Material:** S355 / YSt355; mill certificate required

Size: 70 x 70 x 10 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.375 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	0	12	12.5 through

Final bore/ear spacing subject to current OEM eye drawing. Pair-bore ears after welding; no forced pin alignment.

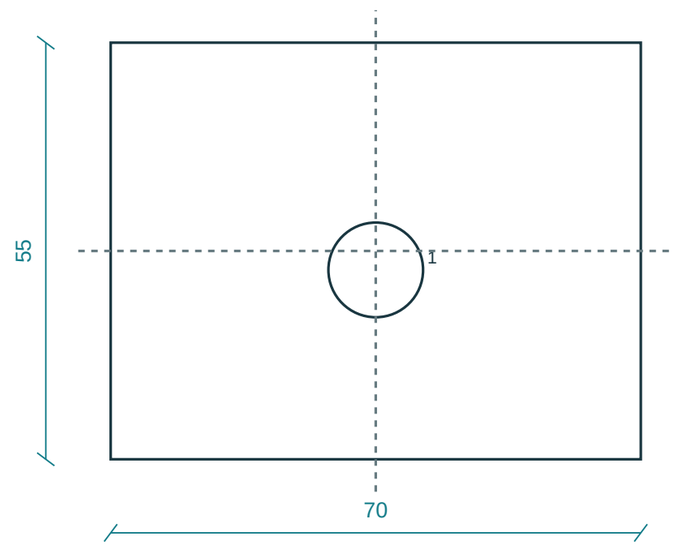
DXF: cad/generated/dxf/P05.dxf | STEP: cad/generated/parts/P05.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P06 / Upper clevis ear

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



Quantity: 2 | **Material:** S355 / YSt355; mill certificate required

Size: 70 x 55 x 10 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.293 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	0	-2.5	12.5 through

Final bore/ear spacing subject to current OEM eye drawing. Pair-bore ears after welding; no forced pin alignment.

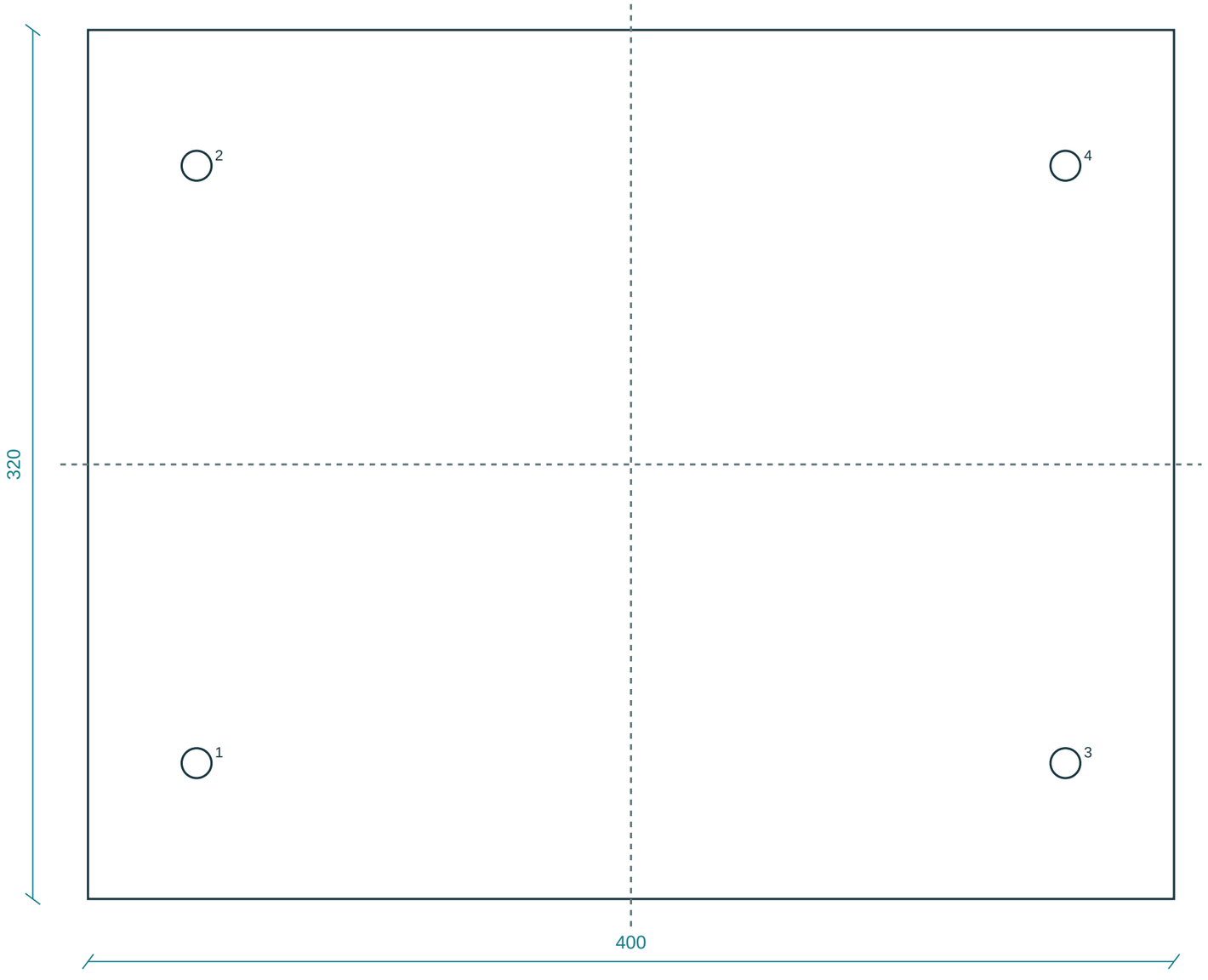
DXF: cad/generated/dxf/P06.dxf | STEP: cad/generated/parts/P06.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P07 / Ballast pan

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



Quantity: 1 | **Material:** S355 / YSt355; mill certificate required

Size: 400 x 320 x 6 thick. Thickness extrudes local +Z.

Nominal unit mass: 6.011 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-160	-110	11 through
2	-160	110	11 through
3	160	-110	11 through
4	160	110	11 through

Ballast pan only. Four M10 through-bolt restraint eyes with reviewed backing/washers/nuts. No human seat or sling attachment rating.

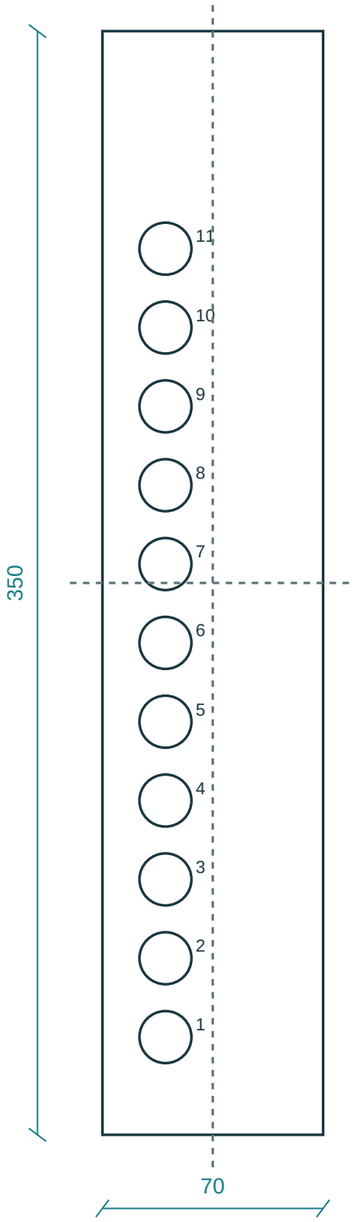
DXF: cad/generated/dxf/P07.dxf | STEP: cad/generated/parts/P07.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P08 / Perch prop ladder plate

Dimensioned engineering review | secured ballast only

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre



Quantity: 4 | **Material:** S355 / YSt355; mill certificate required

Size: 70 x 350 x 10 thick. Thickness extrudes local +Z.

Nominal unit mass: 1.739 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-15	-144	16.5 through
2	-15	-119	16.5 through
3	-15	-94	16.5 through
4	-15	-69	16.5 through
5	-15	-44	16.5 through
6	-15	-19	16.5 through
7	-15	6	16.5 through
8	-15	31	16.5 through
9	-15	56	16.5 through
10	-15	81	16.5 through
11	-15	106	16.5 through

DXF: cad/generated/dxf/P08.dxf | STEP: cad/generated/parts/P08.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P09 / Moving prop contact tab

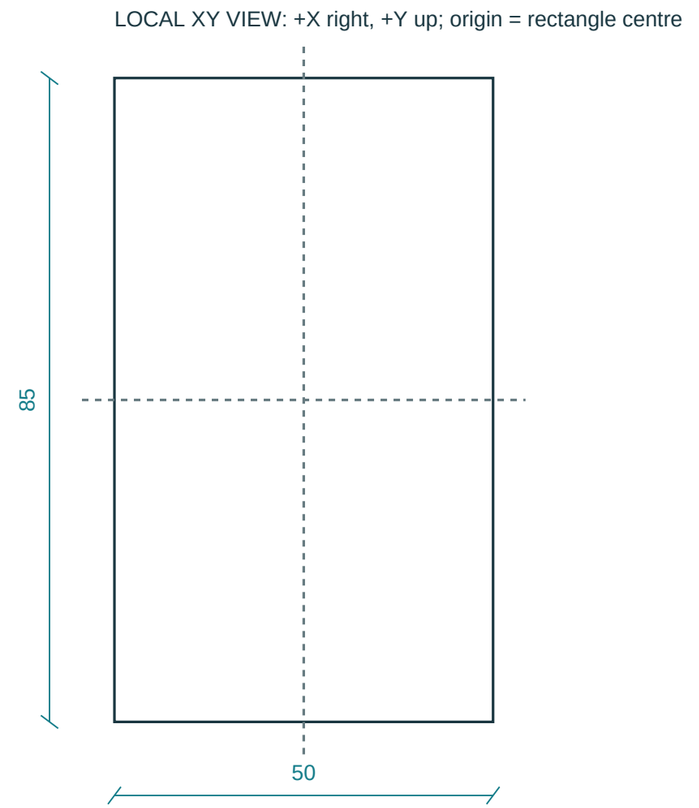
Dimensioned engineering review | secured ballast only

Quantity: 2 | **Material:** S355 / YSt355; mill certificate required

Size: 50 x 85 x 16 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.534 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

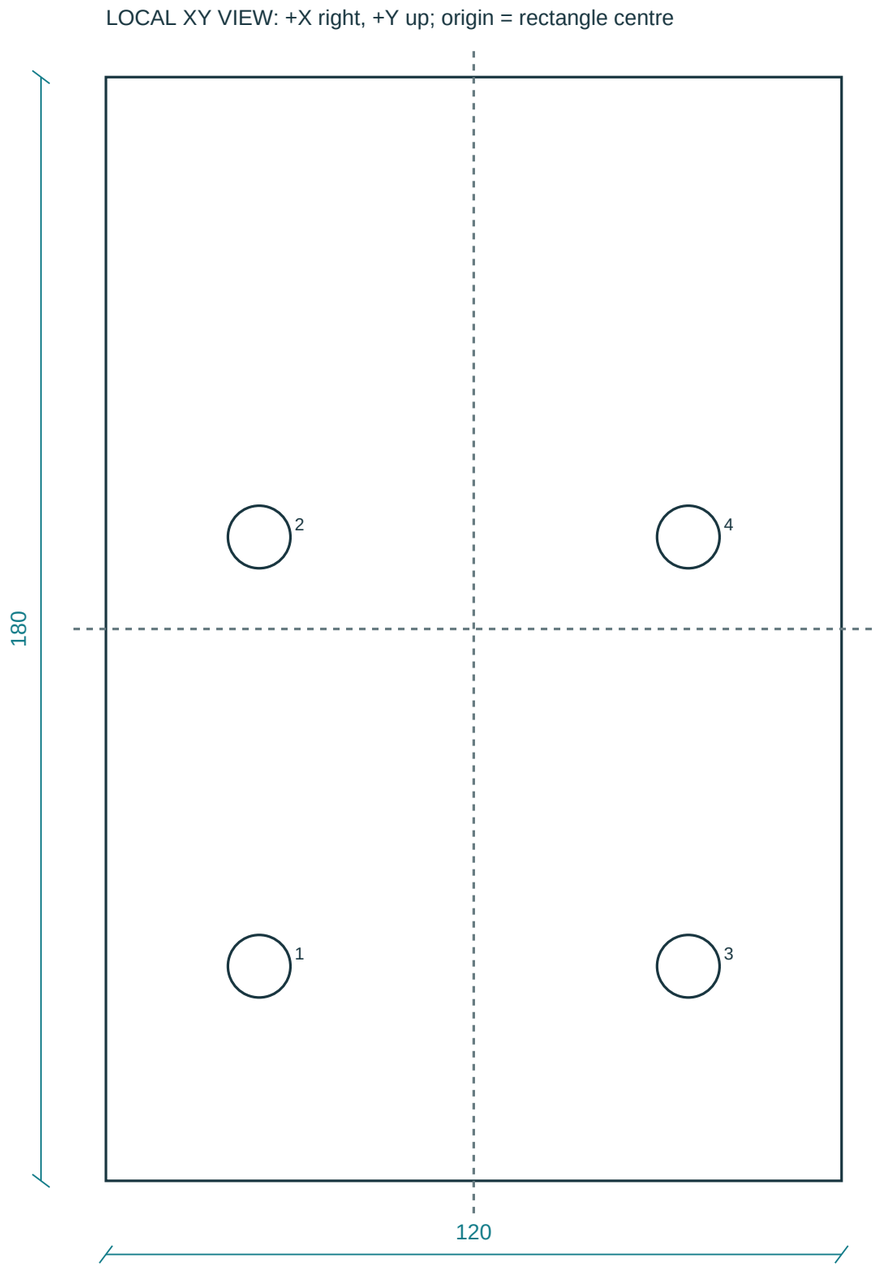


DXF: cad/generated/dxf/P09.dxf | STEP: cad/generated/parts/P09.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P10 / Lower clevis pedestal landing

Dimensioned engineering review | secured ballast only



Quantity: 1 | **Material:** S355 / YSt355; mill certificate required

Size: 120 x 180 x 20 thick. Thickness extrudes local +Z.

Nominal unit mass: 3.340 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

#	X	Y	Dia / operation
1	-35	-55	M12 tap pilot10.2
2	-35	15	M12 tap pilot10.2
3	35	-55	M12 tap pilot10.2
4	35	15	M12 tap pilot10.2

DXF: cad/generated/dxf/P10.dxf | STEP: cad/generated/parts/P10.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P11 / Ladder-to-post bridge

Dimensioned engineering review | secured ballast only

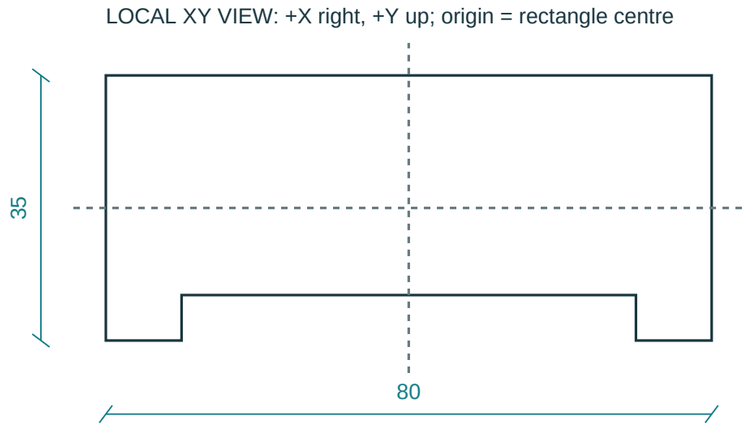
Quantity: 4 | **Material:** S355 / YSt355; mill certificate required

Size: 80 x 35 x 12 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.230 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

MANDATORY centred front-edge relief:60 wide x6 deep through thickness12. Front is local -Y. Do not fill notch with weld.



DXF: cad/generated/dxf/P11.dxf | STEP: cad/generated/parts/P11.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P12 / Moving yoke spacer to bridge

Dimensioned engineering review | secured ballast only

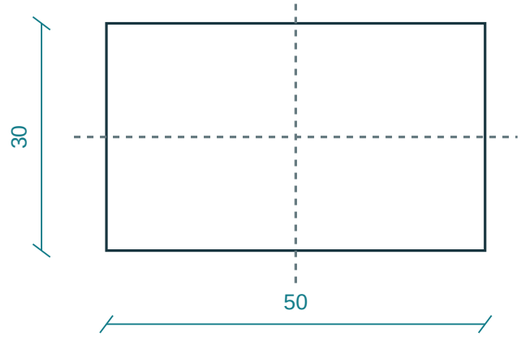
Quantity: 2 | **Material:** S355 / YSt355; mill certificate required

Size: 50 x 30 x 30 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.353 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

LOCAL XY VIEW: +X right, +Y up; origin = rectangle centre

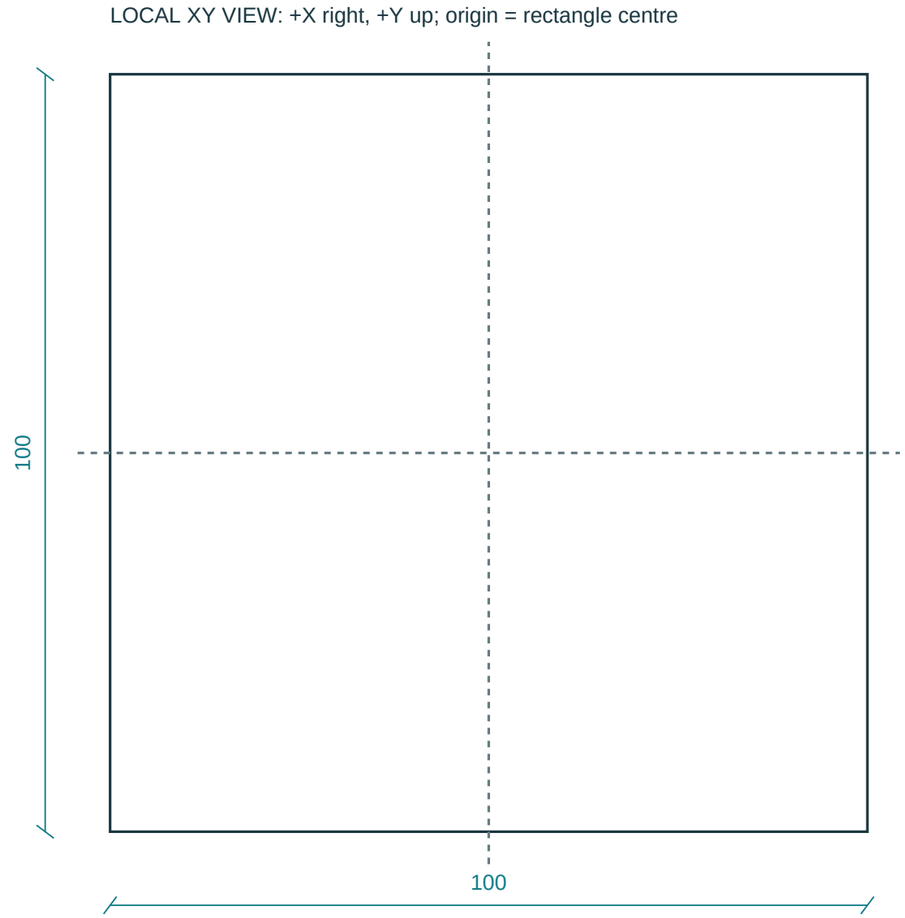


DXF: cad/generated/dxf/P12.dxf | STEP: cad/generated/parts/P12.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P13 / Square mast/base web

Dimensioned engineering review | secured ballast only



Quantity: 2 | **Material:** S355 / YSt355; mill certificate required

Size: 100 x 100 x 6 thick. Thickness extrudes local +Z.

Nominal unit mass: 0.471 kg; sharp-corner/nominal geometry.

Manufacturing: dimensions in mm. General plate +/-0.3; hole positions +/-0.2 unless tighter vendor requirement. Pair-bore mating plates; deburr. Not a weld release.

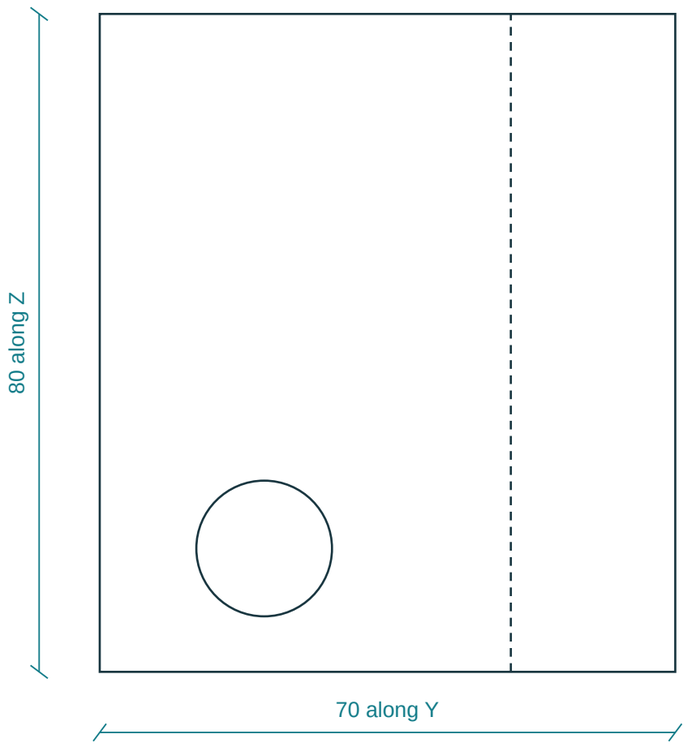
DXF: cad/generated/dxf/P13.dxf | STEP: cad/generated/parts/P13.step

REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P15 / Prop screw block machining

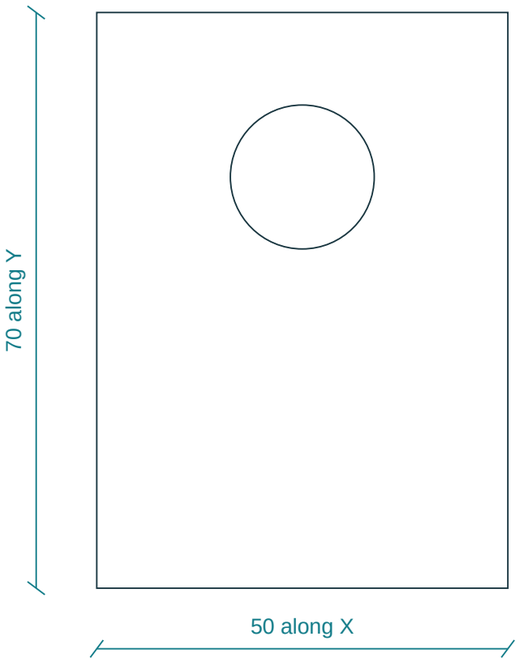
Dimensioned engineering review | secured ballast only

SIDE YZ (pin hole axis X)



Circle centre: Y-15 from part centre, Z15 from underside

TOP XY (thread axis Z)



Qty2; certified S355. Local XY origin at rectangle centre; bottomZ0.

Thread: drill17.5 then tap M20x2.5-6H THROUGH80. Gauge both ends. Do not assemble bolt into an untapped CAD pilot.

Pin: bore16.5 through X at localY-15/Z15. Axis separation30 from thread; nominal ligament11.75. No sharp internal burrs.

Mate: full-thread ISO4017 M20x100-8.8, entered from below. Screw head remains accessible. Apply engineer-approved thread lubricant; record condition.

Hold: full-load thread/pin bearing and eccentric compression review H07. No safety factor or certified capacity assigned.

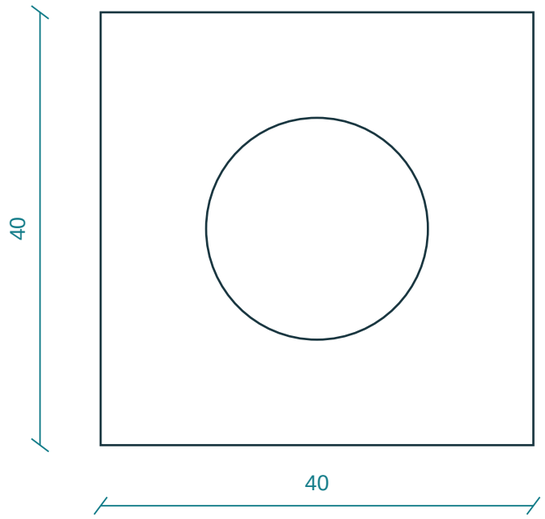
REVIEW ISSUE — NO HUMAN USE — NO FABRICATION OR POWERED-TEST RELEASE

P16 / Prop cap and P17 / Rail retention stop

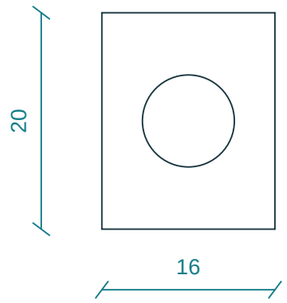
Dimensioned engineering review | secured ballast only

Part	Qty / blank	Machining and assembly
P16	2 /40x40x20	Underside pocket diameter20.5 x3 deep, centred. No through hole or thread. Top face contacts P09; slack non-load-bearing tether.
P17	4 /16X x36Y x20Z	Through bore8.5 parallel Y at X0/Z10; local origin XY centre, bottomZ0. Two at each rail-end working limit.
P17 installation	M8x55 tentative	Backing P02 tap M8-6H at world X+/-26 and Z650/1308. Stop bodies bottomZ640/1298, Y140..176. Nominal2mm gap to blocks at operational ends.

P16 UNDERSIDE; pocket depth3



P17 XZ; bore through Y36



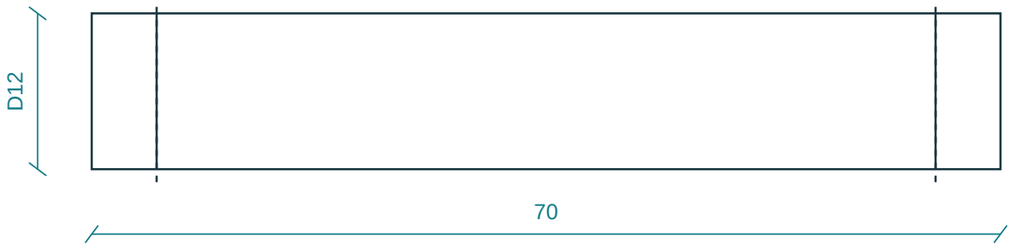
Retention stops limit guide overtravel; they are not qualified fall-arrest devices. Locate hardwired limits to stop before contact. Their bolt/thread/backing/dynamic strength is H08. Screw length includes actual washer,36mm stop and16mm backing engagement; verify no bottoming.

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Pins / Material, bores and positive retention

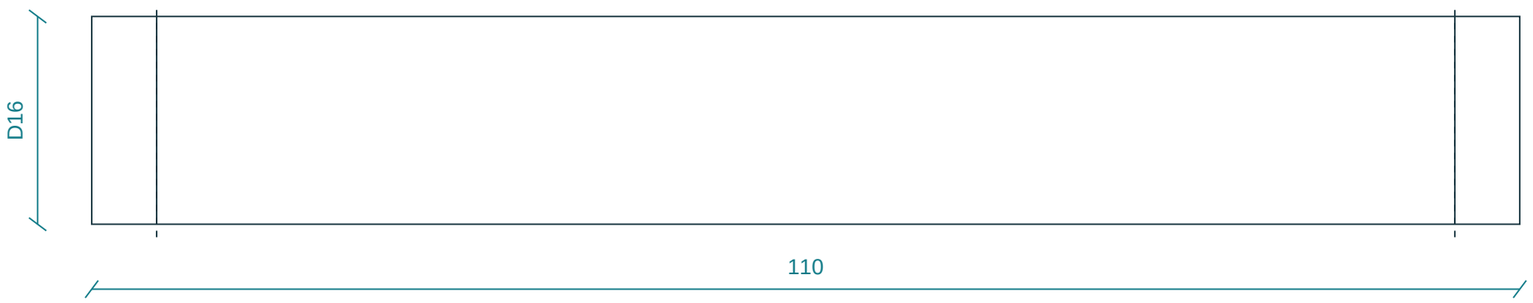
Dimensioned engineering review | secured ballast only

PIN12 / qty2



EN8/C45 certified. Cross drill diameter3 at5 from EACH end, through diameter. Deburr bores. Retain BOTH ends using new3x25 split pins and flat washers. Nominal clearance mating bore12.5 or16.5; engineer approve fit. Pins are not threaded bolts.

PIN16 / qty2



EN8/C45 certified. Cross drill diameter3 at5 from EACH end, through diameter. Deburr bores. Retain BOTH ends using new3x25 split pins and flat washers. Nominal clearance mating bore12.5 or16.5; engineer approve fit. Pins are not threaded bolts.

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Stock / Tube cutlist and first-fit nesting

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ID	Qty	RHS outside / wall	Length	Description
T01	2	40 x 80 x 4	1100	Base longitudinal leg
T02	1	80 x 40 x 4	760	Front base cross member
T03	2	80 x 40 x 4	1200	Mast post
T04	1	60 x 60 x 4	580	Moving front cross member
T10	1	60 x 60 x 4	460	Moving middle cross member; fits between continuous side arms
T05	2	60 x 60 x 4	790	Moving side arm; square-cut butt joints
T06	4	60 x 60 x 4	460	Ballast support drop
T07	2	60 x 60 x 4	700	Ballast cross member
T08	2	60 x 60 x 4	500	Stationary prop support post
T09	2	60 x 60 x 4	100	Prop-post base outrigger

All cuts square at both ends. Same80x40 section may rotate in assembly. First-fit plan uses6000mm bars,20mm trim EACH end and3mm kerf per cut; real supplier length/kerf may differ.

Bar [40, 80, 4] #1: T03:1200, T03:1200, T01:1100, T01:1100, T02:760 | usable offcut 585 mm

Bar [60, 60, 4] #1: T05:790, T05:790, T07:700, T07:700, T04:580, T08:500, T08:500, T10:460, T06:460, T09:100, T09:100 | usable offcut 247 mm

Bar [60, 60, 4] #2: T06:460, T06:460, T06:460 | usable offcut 4571 mm

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Review / Screening calculations and limitations

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Screen	Result / assumption
Payload study	75kg added ballast; not a working-load rating
Moving gravity force	1308N including nominal moving steel, excluding vendor moving mass
Arm beam bending	27.5MPa gravity; 168.3MPa at8000N blocked-drive screen;660mm arm, equal sharing
Guide couple per block	2877N gravity; 17600N fault;300mm spacing, compare C0=52820N with combined loads
Pins nominal double shear	PIN12 35.4MPa; PIN16 19.9MPa,8000N each; local bearing omitted
Caster screen	Steel+75kg /3 = 75.7kg per caster; catalogue110kg each; actual unequal distribution and supplier service factors pending

- Nominal sharp-corner tubes: real sections/corner radii/tolerances need mill properties
- No weld/bolt/thread-stripping/local bearing/buckling/plate bending/fatigue calculation
- Guide load distribution and combined moments/preload/life need supplier review
- Steel-only mass screen excludes vendor parts, guards and welds; stability must use measured contacts/CG
- No patient rating, floor friction/sliding test, proof load or physical result implied
- Swept solid check excludes manually removed props, vendor internals, cables, guards, fasteners and deformation

Passing geometric assertions is not a structural pass. Use reports/checks.json and tests/check_design.py to reproduce values. No physical measurements or tests were performed.

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Release / Assembly, procurement and bench gates

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Hold	Evidence required before proceeding
H01-H04	Current OEM lift code/interfaces; caster revision; genuine matched guide set and machining requirements
H05-H08	Certified material, full structural/weld/joint review, sensor and proof brackets, one-prop retention and independent stop engineering
H09-H10	Final terminal schematic/IO/harness and independent stop tests; guarded cell, calibrated instruments, recovery rig and staged release

Assembly: fixed weldment -> finish-machine guide face -> moving frame -> guided dry sweep -> qualified endstops -> OEM actuator/pins -> props -> ballast restraints/sensors -> guarded cell -> signed electrical integration. Full joint schedule W01-W17 in ASSEMBLY_PLAN.md.

India procurement:60 rows in INDIA_BOM_SOURCING.csv. Four dated-price lines total INR16413.15 with mixed tax bases; this is an incomplete subtotal. OEM kit, fabrication, guarding, integration and test facility are unquoted. Stock listings are not reservations.

Physical stages: lamps and fault injection first; then no added ballast; then proposed5/10/25/50/75kg only after reviewer release. Test stop travel, sensor faults, perch dwell, power loss, emergency recovery and floor stability. No physical cases have passed.

Not included as released detail: human seat/support, approved electrical connector pins, IO PCB, cable routing, final guard cell, match-drilled sensor/cam brackets, joint torques or weld throat sizing. These are explicit holds, not optional fabrication choices.

Source documents: README.md, ASSEMBLY_PLAN.md, ELECTRICAL_INTEGRATION.md, BENCH_TEST_PLAN.md, RELEASE_HOLDS.csv, procurement/INDIA_SELECTION.md and reports/checks.json accompany this PDF.